

$$\int_0^5 \frac{\sqrt{5+x}-\sqrt{5-x}}{\sqrt{5+x}+\sqrt{5-x}} \cdot \frac{1}{\sqrt{25-x^2}} dx$$

$$\int_0^5 \frac{(\sqrt{5+x}-\sqrt{5-x})^2}{(\sqrt{5+x}+\sqrt{5-x})(\sqrt{5+x}-\sqrt{5-x})} \cdot \frac{1}{\sqrt{25-x^2}} dx$$

$$\int_0^5 \frac{5+x-2\sqrt{5+x}\sqrt{5-x}+5-x}{5+x-(5-x)} \cdot \frac{1}{\sqrt{25-x^2}} dx$$

$$\int_0^5 \frac{10-2\sqrt{25-x^2}}{2x} \cdot \frac{1}{\sqrt{25-x^2}} dx$$

$$\int_0^5 \frac{10-2\sqrt{25-x^2}}{2x\sqrt{25-x^2}} dx$$

$$\int_0^5 \frac{5-\sqrt{25-x^2}}{x\sqrt{25-x^2}} dx$$

$$x := 5 \sin \theta \quad dx = 5 \cos \theta d\theta$$

$$\int_0^{\frac{\pi}{2}} \frac{5-5\cos^2 \theta}{5 \sin \theta 5 \cos \theta} 5 \cos \theta d\theta$$

$$\int_0^{\frac{\pi}{2}} \frac{1-\cos \theta}{\sin \theta \cos \theta} \cos \theta d\theta$$

$$\int_0^{\frac{\pi}{2}} \frac{1-\cos \theta}{\sin \theta} d\theta$$

$$\int_0^{\frac{\pi}{2}} \frac{(1-\cos \theta)(1+\cos \theta)}{(\sin \theta)(1+\cos \theta)} d\theta$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^2 \theta}{\sin \theta(1+\cos \theta)} d\theta$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin \theta}{1+\cos \theta} d\theta$$

$$u := 1 + \cos \theta \quad du = -\sin \theta d\theta$$

$$\int_1^2 \frac{1}{u} du$$

$$\ln u \big|_1^2$$

$$\ln 2$$